**Abbreviations**

| | |
|---|---|
| cao | correct answer only |
| dep | dependent |
| FT | follow through after error |
| isw | ignore subsequent working |
| oe | or equivalent |
| SC | Special Case |
| nfww | not from wrong working |
| soi | seen or implied |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | (–3, 7) | **2** | **B1** for correct diagram<br>or correct coordinates for *their* point *D*<br>or for $(-3,k)$ or $(k,7)$ |
| 2(a) | 0.4 oe | **1** | |
| 2(b) | 42<br>0.2 0.2 | **2** | **B1** for 42<br>**B1** for 0.2 and 0.2<br>If B0 scored **SC1** for *their* two probabilities being half *their* **(a)** |
| 3(a) | 40<br>–275 | **2** | **B1** for each |
| 3(b) | 24 | **2** | **B1** for 324 or 289 or $\sqrt{300}$ or 17.3… |
| 4 | 35 | **2** | **B1** for answer 5, 7 or 70<br>or **M1** for 2 × 2 × 5 × 7 and 2 × 3 × 5 × 7<br>or two correct factor trees or tables<br>or 5 × 7 × *k* seen |
| 5(a) | 0.6 or $\frac{3}{5}$ | **1** | |
| 5(b) | 1024 | **1** | |
| 6 | 145 | **3** | **M1** for 180 ÷ 6<br>or any angle congruent to *BCD* = 30<br>**M1** for (360 – 40 – *their* 30) ÷ 2 oe |
| 7(a) | 146 or 146.2 to 146.3 | **3** | **M1** for $\frac{1}{2}\times 12.8\times 12.8$<br>**M1** for $\left[\frac{1}{2}\times\right]\pi\times\left(\frac{12.8}{2}\right)^2$ |

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| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 7(b) | 51[.0] or 51.00 to 51.01… | **4** | **M1** for $\frac{1}{2}\times\pi\times 12.8$<br><br>**M2** for $\sqrt{12.8^2+12.8^2}$ or $\frac{12.8}{\sin 45}$ oe<br>or **M1** for $12.8^2+12.8^2$ oe<br>or $\sin 45=\frac{12.8}{KL}$ oe |
| 8 | $7n + 4$ oe final answer | **2** | **B1** for $7n + j$ or $kn + 4$ $k \neq 0$ ,<br>or $7n + 4$ seen then spoilt |
| 9 | 3375 | **2** | **M1** for $8000\times\left(1-\frac{25}{100}\right)^3$ oe |
| 10 | 1.25 or 1.250… | **3** | **M2** for $\sqrt[8]{\frac{1656.73}{1500}}$ oe<br><br>or **M1** for $1656.73=1500(k)^8$ oe for any $k$ |
| 11 | $y < x$<br>$x < 6$<br>$1 \leqslant y \leqslant 5$ oe | **4** | **B1** for $y < x$<br>**B1** for $x < 6$<br>**B2** for $1 \leqslant y \leqslant 5$<br>or **B1** for $y \geqslant 1$ or $y \leqslant 5$<br><br>If B0 scored,<br>**SC2** for $y \leqslant x$, $x \leqslant 6$ and $1 < y < 5$ oe<br>or **SC1** for three correct from<br>$y = x$, $x = 6$, $y = 1$ and $y = 5$ |
| 12 | Correctly equating one set of coefficients | **M1** | |
| | Correct method to eliminate one variable | **M1** | |
| | $x = 10$, $y = -2$ | **A2** | **A1** for $x = 10$<br>**A1** for $y = -2$<br>If M0 scored **SC1** for 2 values satisfying one of the original equations. |
| 13 | 62 | **3** | **B2** for $m = 20$<br>or **M1** for $5m + 4m = 180$ soi<br>or $p + 4m + 38 = 180$ soi |

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